

Definitions of Functions of Random Variables

1 Functions of One Random Variable

1.1

Theorem 1.1.

For two DRVs X and Y , if $Y = u(X)$ and $X = v(Y)$, where u and v are one-to-one functions, then their pmfs have the following relationship:

$$f_Y(y) = f_X(x) = f_X(v(y))$$

Theorem 1.2.

For two CRVs X and Y , if $Y = u(X)$ and $X = v(Y)$, where u and v are continuous **one-to-one** functions, then their pdfs have the following relationship:

$$f_Y(y) = \frac{f_X(x)}{|dy/dx|} = f_X(v(y)) \left| \frac{d}{dy} v(y) \right| \quad \text{or} \quad f_Y(y)|dy| = f_X(x)|dx|$$

Proof.

By the definition, we have:

$$f_X(x) = \frac{d}{dx} F_X(x) \quad F_X(x) = \mathbb{P}(X \leq x) \quad f_Y(y) = \frac{d}{dy} F_Y(y) \quad F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}[u(X) \leq u(x)]$$

Since u is one-to-one, it is strictly monotonic.

When $u' = dy/dx > 0$,

$$\begin{aligned} F_Y(y) &= \mathbb{P}[u(X) \leq u(x)] = \mathbb{P}(X \leq x) = F_X(x) \\ \implies f_Y(y) &= \frac{d}{dy} F_Y(y) = \frac{d}{dy} F_X(x) = \frac{dx}{dy} \frac{d}{dx} F_X(x) = \frac{f_X(x)}{dy/dx} \end{aligned}$$

When $u' = dy/dx < 0$,

$$\begin{aligned} F_Y(y) &= \mathbb{P}[u(X) \leq u(x)] = \mathbb{P}(X \geq x) = 1 - F_X(x) \\ \implies f_Y(y) &= -\frac{d}{dy} F_X(x) = -\frac{f_X(x)}{dy/dx} \end{aligned}$$

□

Theorem 1.3 (Random Number Generator, RNG).

Let $Y \sim U(0, 1)$ and $F(x)$ have the properties of a cdf of a CRV:

- $F: (a, b) \subseteq \mathbb{R} \rightarrow [0, 1]$ is strictly increasing.
- $F(a) = 0$ and $F(b) = 1$.

Then $X = F^{-1}(Y)$ is a CRV with cdf $F(x)$.

Theorem 1.4.

Suppose X is a CRV with cdf $F_X(x)$ and $Y = F_X(X)$. Then $Y \sim U(0, 1)$.

Proof.

$$\text{Suppose } F(x) = \begin{cases} 0 & x \leq a \\ F(x) & a \leq x \leq b \text{ and } F(x) \text{ is non-decreasing over } (-\infty, \infty). \\ 1 & x \geq b \end{cases}$$

We need to show that $F(x) = F_X(x) = \mathbb{P}(X \leq x)$.

$$\begin{aligned}
 F_X(x) &= \mathbb{P}(X \leq x) = \mathbb{P}(F(X) \leq F(x)) = \mathbb{P}(Y \leq y) \\
 &= F_Y(y) = \begin{cases} 0 & y < 0 \\ y & 0 \leq y \leq 1 \\ 1 & y > 1 \end{cases} \\
 &= \begin{cases} 0 & F(x) < F(a) \\ F(x) & F(a) \leq F(x) \leq F(b) \\ 1 & F(x) > F(b) \end{cases} \\
 &= F(x) \quad \text{for} \quad -\infty < x < \infty
 \end{aligned}$$

□

1.2 Histogram for Continuous Distributions

Histogram gives an approximation of the pdf of a CRV.

The simplest form of a histogram is constructed as follows:

1. Divide the sample space of the distribution into a sequence of adjacent, non-overlapping and equally spaced subintervals.
2. Treat each subinterval as an event. Then count how many observed numerical outcomes fall into each subinterval and calculate the relative frequency
3. Draw a rectangle erected over the bin with height equal to the relative frequency divided by the width of each subinterval.

2 Multivariate Random Variables

Definition 2.1 (Independent).

The n RVs $X_1, X_2, \dots, X_n$ are independent, if

$$f(x_1, x_2, \dots, x_n) = f_{X_1}(x_1)f_{X_2}(x_2) \cdots f_{X_n}(x_n)$$

Remark. A necessary condition for $X_1, X_2, \dots, X_n$ to be independent is that

$$S = S_{X_1} \times S_{X_2} \times \cdots \times S_{X_n}$$

Definition 2.2.

n **independently and identically distributed** (i.i.d.) RVs are called a random sample of size n from a common distribution.

Theorem 2.1.

If $X_1, X_2, \dots, X_n$ are **independent** RVs and $\mathbb{E}[X_i]$ exists for $i = 1, 2, \dots, n$, then

$$\mathbb{E}[X_1 X_2 \cdots X_n] = \mathbb{E}[X_1] \mathbb{E}[X_2] \cdots \mathbb{E}[X_n]$$

Theorem 2.2.

If $X_1, X_2, \dots, X_n$ are **independent** RVs with their variances existing, then

$$\text{Var}(a_1 X_1 + a_2 X_2 + \cdots + a_n X_n) = a_1^2 \text{Var}(X_1) + a_2^2 \text{Var}(X_2) + \cdots + a_n^2 \text{Var}(X_n)$$

Definition 2.3 (Statistic).

Any function of the random sample $X_1, X_2, \dots, X_n$ without any unknown parameters is called a **statistic**.

Definition 2.4 (Sample Mean, or Estimator).

Let $X_1, X_2, \dots, X_n$ be i.i.d. RVs and each X_i follow a distribution with mean μ and variance σ^2 . Then the **sample mean** is defined as

$$\bar{X} := \frac{1}{n} \sum_{i=1}^n X_i$$

It is a **statistic** and also an **estimator** of mean μ .

Theorem 2.3.

Let $X_1, X_2, \dots, X_n$ be i.i.d. RVs and each X_i follow a distribution with mean μ and variance σ^2 . Then

- $\mathbb{E}[\bar{X}] = \mu$
- $\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$

Definition 2.5 ((Adjusted) Sample Variance).

Let $X_1, X_2, \dots, X_n$ be i.i.d. RVs and each X_i follows a distribution with mean μ and variance σ^2 . The **(adjusted) sample variance** is defined as

$$S^2 := \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \quad \text{where} \quad \bar{X} := \frac{1}{n} \sum_{i=1}^n X_i$$

It is a statistic and also an estimator of variance σ^2 .

Remark. The *unadjusted sample variance* is defined as $S^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$, which is biased due to the loss of one degree of freedom. Thus the *unadjusted sample variance* does not converge to the variance σ^2 .

Theorem 2.4.

Let $X_1, X_2, \dots, X_n$ be i.i.d. RVs and each X_i follows a distribution with mean μ and variance σ^2 . Then

- $\mathbb{E}[S^2] = \sigma^2$

Proof.

Given that $\mathbb{E}[\bar{X}] = \mathbb{E}[X_i] = \mu$, $\text{Var}[X_i] = \mathbb{E}[(X_i - \mu)^2] = \sigma^2$, $\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$, we have:

$$\begin{aligned} \sum_{i=1}^n (X_i - \bar{X})^2 &= \sum_{i=1}^n (X_i - \mu + \mu - \bar{X})^2 \\ &= \sum_{i=1}^n (X_i - \mu)^2 + 2(\mu - \bar{X}) \sum_{i=1}^n (X_i - \mu) + \sum_{i=1}^n (\mu - \bar{X})^2 \\ &= \left[\sum_{i=1}^n (X_i - \mu)^2 \right] - n(\mu - \bar{X})^2 \\ \mathbb{E} \left[\sum_{i=1}^n (X_i - \bar{X})^2 \right] &= \sum_{i=1}^n \mathbb{E}[(X_i - \mu)^2] - n \cdot \mathbb{E}[(\mathbb{E}[\bar{X}] - \bar{X})^2] \\ &= n\sigma^2 - n \cdot \frac{\sigma^2}{n} \\ &= (n-1)\sigma^2 \end{aligned}$$

□

3 MGF Technique

Theorem 3.1.

If $X_1, X_2, \dots, X_n$ are **independent** RVs with mgfs $M_{X_i}(t)$, where $|t| < h_i$ for $i = 1, 2, \dots, n$, then the mgf of $Y = \sum_{i=1}^n M_{X_i}(a_i t)$ is given by

$$M_Y(t) = \prod_{i=1}^n M_{X_i}(a_i t)$$

where $|t| < \min\{|h_i/a_i|\}$.

Theorem 3.2.

If $X_1, X_2, \dots, X_n$ are i.i.d. RVs with mgf $M(t)$, where $|t| < h$, then the mgf of $Y = \sum_{i=1}^n X_i$ is given by

$$M_Y(t) = [M(t)]^n \quad \text{for } |t| < h$$

The mgf of the sample mean $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ is given by

$$M_{\bar{X}}(t) = \left[M\left(\frac{t}{n}\right) \right]^n \quad \text{for } \left| \frac{t}{n} \right| < h$$

Theorem 3.3.

The sum of independent chi-squared random variables is itself a chi-squared random variable, with degrees of freedom equal to the sum of the degrees of freedom of the individual variables.

$$X_i \sim \chi^2(r_i) \quad \implies \quad Y = X_1 + X_2 + \dots + X_n \sim \chi^2(r_1 + r_2 + \dots + r_n)$$

Theorem 3.4.

The sum of squares of n independent standard normal RVs is a chi-squared RV with n degrees of freedom.

$$Z_i \sim N(0, 1) \quad \implies \quad Y = \sum_{i=1}^n Z_i^2 \sim \chi^2(n)$$

Proof.

Recall that $Z^2 \sim \chi^2(1)$

□

4 Random Functions Associated with Normal Distributions

4.1

Theorem 4.1.

If $X_i \sim N(\mu, \sigma^2)$ for $i = 1, 2, \dots, n$, then the sample mean $\bar{X}$ satisfies

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

and the sample variance S^2 satisfies

$$\frac{n-1}{\sigma^2} S^2 \sim \chi^2(n-1)$$

Moreover, $\bar{X}$ and S^2 are **independent**, and the normal distribution is the only distribution where they are independent.

Remark. If $X_i \sim N(\mu, \sigma^2)$ for $i = 1, 2, \dots, n$, then

$$\sum_{i=1}^n \frac{(X_i - \mu)^2}{\sigma^2} \sim \chi^2(n) \quad \sum_{i=1}^n \frac{(X_i - \bar{X})^2}{\sigma^2} \sim \chi^2(n-1)$$

When μ is replaced by $\bar{X}$, one degree of freedom is lost.

4.2 Student's t -Distribution

Definition 4.1 (Student's t -Distribution).

Parameter $r = 1, 2, 3, \dots$

The **Student's t -distribution** with r degrees of freedom is defined as

$$T = \frac{Z}{\sqrt{U/r}} \sim t(r)$$

where $Z \sim N(0, 1)$ and $U \sim \chi^2(r)$ are independent.

Theorem 4.2.

The pdf of $T \sim t(r)$ is given by

$$f(t) = \frac{\Gamma(\frac{r+1}{2})}{\sqrt{r\pi} \Gamma(\frac{r}{2})} \left(1 + \frac{t^2}{r}\right)^{-\frac{r+1}{2}} \quad \text{for } -\infty < t < \infty$$

Student's t -distribution is heavier-tailed than the standard normal distribution.

Theorem 4.3.

Let $X_i \sim N(\mu, \sigma^2)$ for $i = 1, 2, \dots, n$ be independent, and let $\bar{X}$ and S^2 be the sample mean and sample variance, respectively. Then,

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1) \quad T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1)$$

5 The Central Limit Theorem

Definition 5.1 (Convergence in Distribution).

A sequence of RVs X_n converges in distribution, or converges weakly, or converges in law to a RV X if

$$\lim_{n \rightarrow \infty} \mathbb{P}(X_n \leq x) = \mathbb{P}(X \leq x) \quad \forall x$$

Denoted as

$$X_n \xrightarrow{d} X$$

Remark. The convergence of RVs can be characterized in multiple ways. Examples include:

- Convergence in Probability
- Convergence in Distribution
- Almost Sure Convergence
- Convergence in L^p Space

Convergence in distribution is the weakest form of them.

Theorem 5.1 (Central Limit Theorem, CLT).

Let $\bar{X}$ be the sample mean of a random sample of size n from a distribution with a finite mean μ and a finite non-zero variance σ^2 . Then,

$$\bar{X} \xrightarrow{d} N\left(\mu, \frac{\sigma^2}{n}\right) \quad \text{as } n \rightarrow \infty$$

Or equivalently,

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1) \quad \text{as } n \rightarrow \infty$$

6 Approximations for Discrete Distributions

We can approximate a discrete distribution with a continuous distribution that has a pdf similar to the histogram of the discrete distribution.

6.1 Half-Unit Correction for Continuity

Let $Y = \sum_{i=1}^n X_i$, where $X_1, \dots, X_n$ are i.i.d. random samples from a discrete distribution with mean μ and variance σ^2 . By the CLT, for large n , Y can be approximated by $Z \sim N(n\mu, n\sigma^2)$.

$$\mathbb{P}(Y = k) = \mathbb{P}\left(k - \frac{1}{2} < Y < k + \frac{1}{2}\right) \approx \mathbb{P}\left(k - \frac{1}{2} < Z < k + \frac{1}{2}\right) = \Phi\left(\frac{k + \frac{1}{2} - n\mu}{\sigma\sqrt{n}}\right) - \Phi\left(\frac{k - \frac{1}{2} - n\mu}{\sigma\sqrt{n}}\right)$$

Remark. We need another CRV as a bridge, say T . T should have a non-continuous pdf that has the same shape as the histogram of the DRV Y . Then we can say that

$$\mathbb{P}\left(k - \frac{1}{2} < Y < k + \frac{1}{2}\right) = \mathbb{P}\left(k - \frac{1}{2} < T < k + \frac{1}{2}\right) \approx \mathbb{P}\left(k - \frac{1}{2} < Z < k + \frac{1}{2}\right)$$

Theorem 6.1.

For large n , the binomial distribution $B(n, p)$ can be approximated by $N(np, np(1-p))$.

Proof.

Let $Y = \sum_{i=1}^n X_i$, where $X_i \sim B(1, p)$ and X_i are i.i.d. Then $Y \sim B(n, p)$, and it can be approximated by $N(np, np(1-p))$. $\square$

Theorem 6.2.

For large enough λ , the Poisson distribution $\text{Poisson}(\lambda)$ can be approximated by $N(\lambda, \lambda)$.

Proof.

Here, provide a proof for λ being an integer. Let $Y = \sum_{i=1}^{\lambda} X_i$, where $X_i \sim \text{Poisson}(1)$ and X_i are i.i.d. Then $Y \sim \text{Poisson}(\lambda)$, and it can be approximated by $N(\lambda, \lambda)$ when λ is large enough, since $\mathbb{E}[X_i] = \text{Var}(X_i) = 1$. $\square$

7 Chebyshev's Inequality and Convergence in Probability

Theorem 7.1 (Markov's Inequality).

If the RV X is **non-negative** and has a finite mean, then,

$$\mathbb{P}(X \geq \varepsilon) \leq \frac{\mathbb{E}[X]}{\varepsilon} \quad \forall \varepsilon > 0$$

Theorem 7.2 (Chebyshev's Inequality).

If the RV X has a finite mean μ and finite nonzero variance σ^2 , then,

$$\mathbb{P}(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2} \quad \forall k > 0$$

Or equivalently,

$$\mathbb{P}(|X - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2} \quad \forall \varepsilon > 0$$

Remark. For any given RV X with mean μ and variance $\sigma^2 > 0$, the corresponding **standardized RV** is $\frac{X - \mu}{\sigma}$, with mean 0 and variance 1.

Proof (Chebyshev's Inequality).

We only need to show that $\mathbb{P}(|Z| \geq k) \leq \frac{1}{k^2}$, where Z is the standardized RV.

$$\mathbb{E}[Z^2] = 1 = \int_{-\infty}^{\infty} z^2 \mathbb{P}(Z = z) \geq \int_{-\infty}^{-k} k^2 \mathbb{P}(Z = z) + 0 + \int_k^{\infty} k^2 \mathbb{P}(Z = z) = k^2 \mathbb{P}(|Z| \geq k)$$

□

Theorem 7.3 (Chebyshev's Inequality).

If the RV X has a finite mean μ and finite nonzero variance σ^2 , then,

$$\mathbb{P}(|X| \geq \varepsilon) \leq \frac{\mathbb{E}[X^2]}{\varepsilon^2} \quad \forall \varepsilon > 0$$

Proof.

When X is not standardized, i.e., $\mathbb{E}[X^2] = \sigma^2 + \mu^2 \neq 1$,

$$\mathbb{E}[X^2] = \int_{-\infty}^{\infty} x^2 \mathbb{P}(X = x) \geq \int_{-\infty}^{-\varepsilon} \varepsilon^2 \mathbb{P}(X = x) + 0 + \int_{\varepsilon}^{\infty} \varepsilon^2 \mathbb{P}(X = x) = \varepsilon^2 \mathbb{P}(|X| \geq \varepsilon)$$

□

Definition 7.1 (Convergence in Probability).

A sequence of RVs $\{X_n\}$ is said to converge in probability to a RV X , and denoted as $X_n \xrightarrow{p} X$, if

$$\lim_{n \rightarrow \infty} \mathbb{P}(|X_n - X| \geq \varepsilon) = 0 \quad \forall \varepsilon > 0$$

Theorem 7.4 (The Weak Law of Large Numbers, WLLN).

Let $X_1, X_2, \dots, X_n$ be i.i.d. RVs with finite mean μ and nonzero variance. Then,

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{p} \mu$$

Proof.

Since $\mathbb{E}[\bar{X}] = \mu$ and $\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$, by Chebyshev's inequality, we have

$$\begin{aligned} \mathbb{P}(|\bar{X} - \mu| \geq \varepsilon) &\leq \frac{\sigma^2}{n\varepsilon^2} \quad \forall \varepsilon > 0 \\ \Rightarrow \lim_{n \rightarrow \infty} \mathbb{P}(|\bar{X} - \mu| \geq \varepsilon) &= 0 \quad \forall \varepsilon > 0 \end{aligned}$$

□

8 Limiting MGF Technique

Theorem 8.1.

If X_n is a sequence of RVs with characteristic function $\varphi_n(t)$, and X is a RV with characteristic function $\varphi(t)$, then

$$\lim_{n \rightarrow \infty} \varphi_n(t) = \varphi(t) \quad \Longleftrightarrow \quad X_n \xrightarrow{d} X$$

Remark. $\varphi(t) := \mathbb{E}[e^{itX}] = M(it)$